

ERC Starting Grant 2026

Part B1

Flexible Intelligence through Neurocomputational Design

FIND

Principal Investigator: Dr. Mikkel Elle Lepperød
Host Institution: Simula Research Laboratory, Norway
Duration: 60 months
Primary Panel: PE6 — Computer Science and Informatics

Abstract

Current AI systems cannot selectively store and retrieve information across thousands of timesteps—a capability biological brains achieve routinely through distributed memory systems. Transformers forget beyond their context window; state space models compress everything into fixed-size state; and memory-augmented networks lack causal models to decide what is worth storing. FIND develops **PAM** (**P**redictive **A**ctive temporal coding with distributed **M**emory), a novel brain-inspired architecture where modules of neurons read and write to local long-term memory, combining temporal predictive coding, hyperdimensional computing, and active inference into a unified framework.

PAM combines three mechanisms: (1) *Distributed modular specialization* where each module maintains dedicated content-addressable memory; (2) *Temporal predictive coding* where prediction errors drive memory updates and learning; and (3) *Active inference* where action selection minimizes uncertainty. Our preliminary results demonstrate PAM’s viability beyond toy benchmarks: 100% recognition accuracy on CORe50 (50 objects) with **zero additional forgetting by construction**—non-parametric storage cannot overwrite parameters; **98.2% online continual learning accuracy** across 120K samples in 55 seconds—12× faster than gradient-based methods; and **24.7% exact match on ARC** (the Abstraction and Reasoning Corpus) with only 1.47M parameters and no pretraining, approximately 2× GPT-4o’s 12% at >1000× fewer parameters.

FIND will develop PAM through three work packages: **WP1** establishes temporal predictive coding with distributed memory for continual learning and long-range temporal reasoning; **WP2** extends this to compositional reasoning and program synthesis through neurosymbolic HDC integration; and **WP3** stress-tests the system in demanding open-ended environments (Memory Maze, Minecraft, Factorio) requiring planning and adaptation across >10,000 timesteps.

FIND: Flexible Intelligence through Neurocomputational Design

1 Part I of the Scientific Proposal

We live in time. Memory bridges past and present, enabling agents to learn from experience and make informed decisions. Humans develop internal world models for planning and reasoning; replicating this in machines is the next step toward flexible AI [1]. Current AI struggles to generalise [2], lacking long-term memory [3], causal understanding [4], and life-long learning without catastrophic forgetting [5].

Consider an agent managing a Factorio production chain (Figure 2). It discovers copper wire requires copper plates (step 2,000) and circuits require wire plus iron (step 8,000), then encounters a blueprint requiring 50 circuits at step 15,000. It must recall the dependency chain across >10,000 timesteps, reason causally about bottlenecks, and adapt its layout. Current systems cannot do this: transformers cannot attend across 10,000 steps [6]; SSMs compress dependencies into fixed-size state [7]; MANNs lack causal models to identify bottlenecks [8]. The challenge is to achieve this *learned from raw experience*, without a pretrained model or hand-engineered memory plumbing. **How can systems intelligently select what to retain, discard, and retrieve to learn across long temporal sequences?**

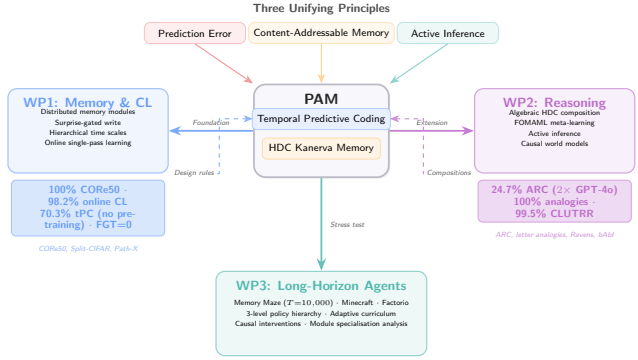


Figure 1: Conceptual overview of FIND. PAM (centre) unifies temporal predictive coding and HDC-based content-addressable memory. Three work packages build outward: **WP1** establishes the memory and continual learning foundation; **WP2** extends with compositional reasoning and causal world models; **WP3** stress-tests in demanding long-horizon environments. Preliminary results (coloured boxes) validate each WP’s core claims.



Figure 2: The capability gap FIND addresses. An agent discovers dependencies thousands of steps apart and must later recall and compose them under a new goal. This requires selective storage, content-addressable recall, and causal reasoning across >10,000 timesteps—beyond current architectures (bottom).

1.1 Ground-breaking nature and impact

FIND proposes a paradigm shift drawing inspiration from biological memory systems. We develop **PAM** (Predictive Active temporal coding with distributed Memory), a flat modular architecture where complexity arises from learned routing and memory interactions, operating through:

- Specialized modules with dedicated Kanerva memory [9, 10]
- HDC symbolic representations [11] retained in memory
- Learned orchestration for dynamic information routing
- Temporal predictions via predictive coding; action selection via active inference [12, 13]

FIND’s novelty lies at the intersection of three established fields in a configuration no prior system has achieved simultaneously:

- **Temporal predictive coding** [14] with **distributed external memory**—not just recurrent state: prior tPC models store context solely in hidden activations; FIND externalises memory into content-addressable HDC slots persistent across thousands of timesteps.
- **Memory-augmented networks** [8] with **biologically-grounded iterative inference**—not single-pass read/write: prior MANNs use a single forward pass to address memory; FIND uses

iterative predictive-coding inference to resolve ambiguous queries, yielding a $1.75\times$ recall advantage at $\sigma=0.4$.

- **Hyperdimensional computing [11] embedded as the storage format of a gradient-trained system**—not as a standalone symbolic method: prior HDC systems operate outside neural training loops, and even NVSA [15], which couples vector-symbolic reasoning to a trained perception network, is a single-task inference pipeline without persistent temporal memory, continual learning, or action; FIND makes HDC the storage substrate of a continual, temporal system, integrating binding directly into the memory update.

Table 1 situates FIND against the state of the art. No existing system combines all four capabilities.

Table 1: Positioning of PAM against existing approaches. Only PAM combines external distributed memory, iterative inference, symbolic binding, and active inference in a single end-to-end trainable architecture.

System	External Memory	Iterative Inference	Symbolic Binding	Active Inference
NTM / DNC [16, 8]	✓	—	—	—
Kanerva Machine [9]	✓	—	—	—
Temporal PC [14]	—	✓	—	—
HDC systems [11]	—	—	✓	—
S4 / Mamba [7]	— (compressed state)	—	—	—
Titans [17]	✓	—	—	—
NVSA [15]	—	—	✓	—
Active PC [13]	—	✓	—	✓
PAM (FIND)	✓	✓	✓	✓

1.2 State of the art, knowledge needs, and project objectives

Temporal memory and predictive coding

A key challenge in AI is creating memory systems comparable to the brain. RAG [18, 19] and agentic systems like MemGPT [20] manage context through external plumbing, but these sidestep bottom-up lifelong learning and struggle with out-of-distribution scenarios [2].

The core challenge is separating memory storage from computation [21, 22]. MANNs—NTMs [16], DNCs [8], Kanerva Machines [9]—address this partially. HDC extends sparse distributed memories with symbolic binding [11, 23]. Long-context transformers (Infini-Attention [24]) extend windows but do not solve selective storage. Modern SSMs—Mamba-2 [25], xLSTM [26]—achieve efficient sequence processing but compress into fixed-size state.

Predictive coding (PC) minimizes prediction errors between expectations and inputs [27, 12]. Extensions to active [13] and temporal PC [14] enable biologically plausible cognitive maps. Test-time training (TTT) [28] shares PAM’s iterative adaptation principle but modifies the full model rather than memory slots. Titans [17] is closest in spirit: it gates test-time writes to a long-term memory by surprise—sharing PAM’s write-gating principle—but stores into a monolithic parametric memory without symbolic binding, compositional retrieval, or action selection. **However, current PC models rely on Markov assumptions, unable to handle long-term credit assignment.**

Causal reasoning, world models, and compositional understanding

A causal world model represents generative mechanisms enabling counterfactual reasoning [29, 4]. Recent models—JEPA [30], DINO-WM [31], GAIA-1 [32]—show promise, but current benchmarks operate at irrelevant timescales: 2000 steps [33] vs. desert ants maintaining memories across 200 000 steps [34].

A complementary gap is *compositional reasoning*: factorizing knowledge into reusable subspaces for novel recombinations. The ARC benchmark [35] exemplifies this—tasks require recognising that transformations *compose* algebraically. FIND addresses this through HDC binding (bind, unbind, superposition) within the same space used for memory.

Cognitive map learning has been pursued in neuroscience [36] and AI [37], including by our lab [38, 39, 40, 41], but these lack a unified theory of perception, cognition, and action—the gap FIND addresses.

Adaptive learning

Life-long learning requires managing information without catastrophic forgetting. Current approaches span regularization, replay, and Bayesian methods [5, 42]; we have explored generative stability [43] and plasticity [44]. FIND addresses this through HDC [11]: binding memories as compositional vectors enables fast retrieval and symbolic queries without retraining.

1.3 Research strategy and objectives

Central hypothesis: Predictive coding over content-addressable distributed memory slots is sufficient for selective long-horizon recall—the ability to store, retrieve, and compose information across thousands of timesteps without catastrophic forgetting—and achieves it where compressed-state methods cannot without unbounded state growth. We decompose this into three testable sub-hypotheses:

- **H1 (WP1):** Non-parametric HDC-structured memory is immune to parameter-overwriting forgetting *by construction*—a guarantee gradient-based methods do not offer—while maintaining competitive recognition accuracy; our CORE50 results (100% recognition, 98.2% online CL accuracy) confirm the accuracy side.
- **H2 (WP2):** Compositional binding in HDC space enables zero-shot generalisation to novel transformation compositions, with distributed memory as a load-bearing component of test-time program synthesis—PAMNCA ablations attribute 30–43% of performance to memory, growing with compute budget.
- **H3 (WP3):** Memory-augmented agents outperform compressed-state agents (SSMs, Transformers) specifically on tasks requiring recall across >5,000 timesteps.

We address this through three objectives and corresponding work packages:

- **Objective 1 / WP1: Temporal Predictive Coding with Distributed Memory.** Develop the foundational PAM architecture that processes information over different time scales through temporal predictive coding with distributed memory modules.
- **Objective 2 / WP2: Causal Reasoning and Compositional Understanding.** Extend PAM with active inference, neurosymbolic integration, and subspace-based compositional reasoning for causal world model learning.
- **Objective 3 / WP3: Long-Horizon Adaptation in Open-Ended Environments.** Stress-test the complete PAM system in demanding environments (mazes, Minecraft, Factorio) that require sophisticated planning, resource management, and long-horizon adaptation.

Theoretical approach. PAM combines three mechanisms:

1. *Distributed modular specialization:* Specialized modules with dedicated Kanerva memory [9], routed via Gumbel-softmax program induction iteratively over n stages.
2. *Temporal predictive coding:* Prediction errors drive memory updates and model learning.
3. *Active inference:* Action selection minimizes uncertainty, enabling efficient exploration and causally relevant memory storage.

PAM uses content-addressable memory with retrieval via similarity between z_t^m and stored basis vectors h_i^m , with exponential maps [45] for closed-form Riemannian updates. HDC [11, 45] bridges neural and symbolic approaches, enabling compositional reasoning through binding operations (Figure 3).

Preliminary results. The BioAI group has developed FIND’s foundations through neurocomputational memory research: grid/place/border cells emerge spontaneously in AI agents [38], orthogonal remapping encodes context [40, 41, 39], NCA enables self-organizing module communication [46], one-shot memory transfer [47], and abstract reasoning [48]. The tPC-KM system achieves 98.1% exact recall at $\sigma=0.4$ ($1.75\times$ over feedforward) and $3.75\times$ less forgetting than tPC-alone (Figure 4).

WP1: Memory and continual learning. Beyond the MNIST benchmark, PAM achieves **100% recognition accuracy** on CORE50 (50 real-world objects, $T=10,000$ steps) with both pretrained and raw pixel features. On the official CORE50-NI online continual learning protocol (test-then-train, single pass, no task boundaries), PAM achieves **98.2% online accuracy** across 8 experiences (120K samples) in 55 seconds— $12\times$ faster than gradient-based replay methods—with positive backward transfer and *zero additional forgetting* as experiences accumulate: non-parametric storage cannot overwrite parameters, so accuracy is bounded only by superposition capacity, which scales predictably with memory dimension [10]. With PAM’s own tPC encoder (zero pretraining), the system reaches **70.3% per-experience accuracy** while learning both representations and memory online—the only

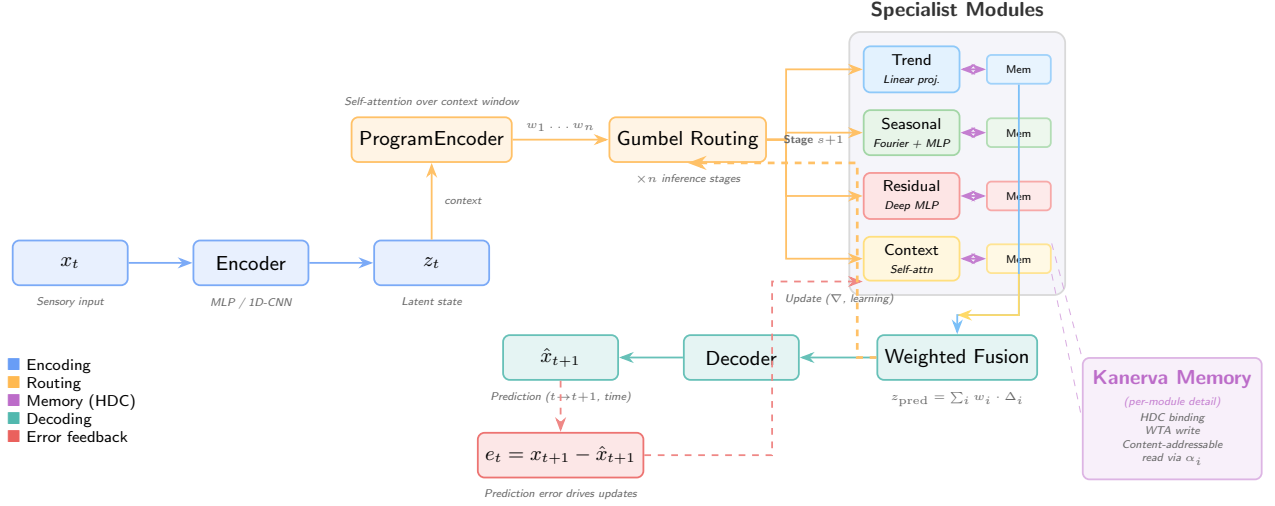


Figure 3: PAM architecture. Sensory input x_t is encoded into latent state z_t . A ProgramEncoder infers routing weights via self-attention over the context window. Gumbel-softmax routing selects among four specialist modules (Trend, Seasonal, Residual, Context), each with dedicated Kanerva memory using HDC binding for content-addressable read/write. The routing is applied iteratively over n stages. Module outputs are fused into a prediction $\hat{x}_{t+1}$; prediction errors drive parameter updates. The module inventory is task-dependent (shown here for temporal forecasting).

continual learning system that operates without a pretrained backbone. The gap to the frozen-feature setting reflects *representation drift*: online-learned keys diverge from those stored earlier, degrading content-addressable retrieval—the central open problem for episodic memories keyed by learned embeddings, which WP1 addresses with a key-reconsolidation mechanism.

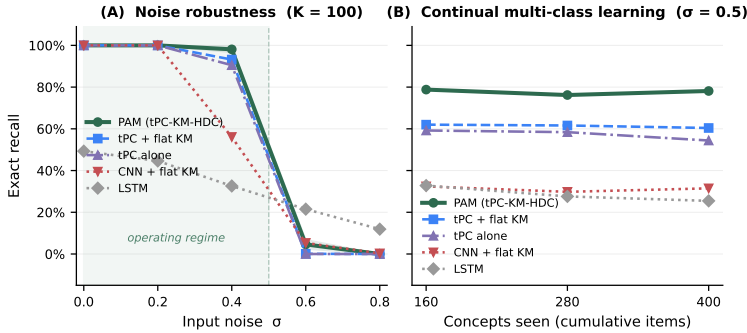


Figure 4: Preliminary PAM results (WP1 foundation). (A) Exact recall vs. input noise σ ($K=100$ items, 20 trials). Iterative inference (PAM, tPC-KM) outperforms feedforward encoding (CNN-KM) by $1.75\times$ at $\sigma=0.4$. Beyond $\sigma\approx 0.5$ (shaded band), exact recall collapses for all content-addressable methods—a capacity/SNR phase transition predicted by HDC crosstalk analysis, since crosstalk in superposed hypervectors grows with stored items; deriving this boundary analytically is a T1.2 target (Part II). (B) Exact recall across three sequential class phases ($\sigma=0.5$). PAM maintains $\approx 78\%$ recall across all phases; HDC binding prevents inter-class interference as new concepts accumulate, reducing forgetting $3.75\times$ versus tPC alone.

tribution *growing* with compute budget. FOMAML meta-learning provides a $2.5\times$ effective compute speedup, validating T2.2’s meta-learning approach. Separately, our HDC system achieves 100% on direct analogies (with 200 distractors), 99.5% on CLUTRR 2-hop relational reasoning, and 100% on Raven’s Progressive Matrices (NVSA [15] reaches similar accuracy on Raven’s; PAM’s distinction is that the same memory substrate also serves continual temporal recall rather than a dedicated reasoning pipeline).

PAM makes two testable predictions unique to its architecture: (1) non-parametric memory guarantees zero *additional* forgetting by construction (recall bounded only by superposition capacity) while

WP2: Compositional reasoning.

We validated PAM’s program synthesis capabilities on the Abstraction and Reasoning Corpus (ARC) [35]. Our PAMNCA system—a neural cellular automaton with distributed PAM memory (1.47M parameters)—achieves **24.7% exact match** (321/1302 episodes) through test-time training, with no pretrained backbone. The significance is the efficiency frontier, not the absolute score: large program-synthesis ensembles now exceed this number using 10^5 – $10^6\times$ more compute, whereas PAMNCA doubles GPT-4o’s 12% at $>1000\times$ fewer parameters, learning each task from its support examples alone. Memory ablations show the distributed memory module contributes 30–43% of performance, with its contri-

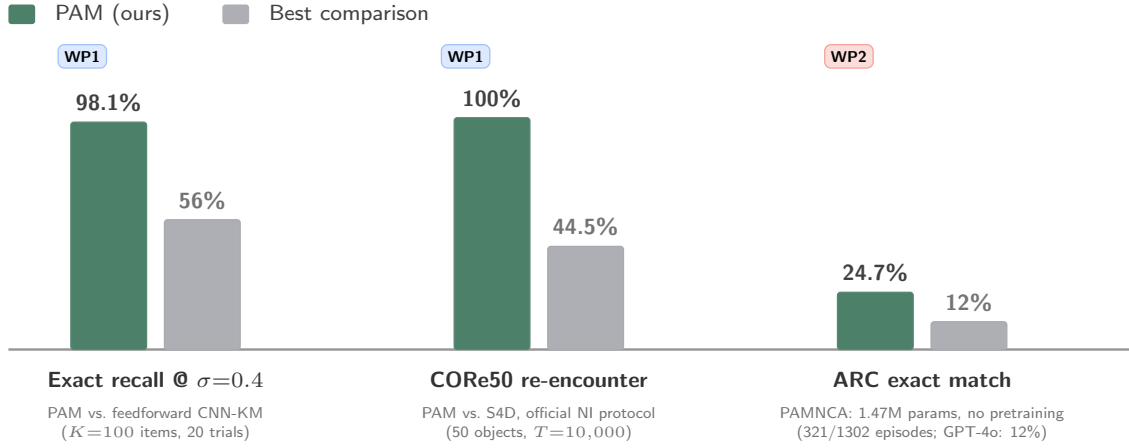


Figure 5: Headline preliminary results. PAM against the strongest comparison on each benchmark: noisy sequence recall at $\sigma=0.4$ (WP1 foundation), CORE50 re-encounter recognition after $T=10,000$ steps (WP1), and ARC program synthesis via test-time training (WP2).

maintaining recognition accuracy—our CORE50 experiments confirm 100% recognition versus the S4D baseline’s 44.5% re-encounter accuracy; (2) distributed memory is load-bearing for program synthesis via test-time adaptation, with its contribution growing with task complexity—our PAMNCA ablations support this (30% at TTT=500, 43% at TTT=1000).

1.4 Major expected outcomes

Science: A theoretical framework unifying predictive coding, distributed memory, and active inference beyond Markov assumptions. Understanding of how modular specialization emerges from predictive principles.

Technology: PAM as open-source framework with benchmarks spanning forecasting (ETT, Weather), long-range memory (Path-X, Memory Maze), and continual learning. Orchestration algorithms with reproducible evaluation suites.

Society: Foundations for AI with principled memory applicable to healthcare (patient timelines), industrial automation (Factorio-style planning), and environmental monitoring.

Success criteria: (i) $\geq 90\%$ recall at $\sigma=0.3$ and MSE within 10% of PatchTST (WP1); (ii) $>40\%$ ARC exact match at $<10\text{M}$ parameters without a pretrained backbone, extending current 24.7% (WP2); (iii) >10 pp over DQN on Memory Maze at $T>5,000$ (WP3). Team: PI (50%), one postdoc (ML systems), two PhDs; 4-tier benchmark hierarchy (Part II).

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2 Curriculum Vitae and Track Record

Personal details

Name: Dr. Mikkel Elle Lepperød
Date of birth: 14 September 1985
Nationality: Norwegian
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Education and key qualifications

2020 **Ph.D.** Neuroscience, Faculty of Medicine, University of Oslo, Norway
2014 **M.Sc.** Applied Mathematics, Norwegian University of Life Sciences, Norway

Current and relevant previous positions

2026– **Senior Research Scientist** (equiv. associate professor), Simula Research Laboratory
2023– **Adjunct Associate Professor** (20%), Dept. of Physics, University of Oslo
2022–2026 **Research Scientist** (equiv. assistant professor), Simula Research Laboratory
2020–2022 **Postdoctoral Fellow**, Dept. of Biosciences, University of Oslo

PhD supervisor: Prof. Torkel Hafting and Prof. Marianne Fyhn (University of Oslo).

Postdoctoral mentor: Self-directed (group leader from 2020).

Career breaks: Parental leave (5 months in 2017, 5 months in 2020).

Research outputs

The following ten outputs demonstrate my trajectory from experimental neuroscience to neuro-inspired AI, establishing the foundations for FIND:

1. **Lepperød ME** et al. *Optogenetic pacing of medial septum parvalbumin-positive cells disrupts temporal but not spatial firing in grid cells*. Science Advances, 2021. Role: First author, conceived and led the study. Significance: Established causal evidence for temporal coding in the entorhinal cortex, directly motivating the temporal predictive coding component of PAM.
2. **Lepperød ME** et al. *Inferring causal connectivity from pairwise recordings and optogenetics*. PLoS Computational Biology, 2022. (With K. Kording.) Role: First author. Significance: Developed quasi-experimental methods for causal inference in neural circuits, grounding FIND’s causal reasoning objectives.
3. Schøyen VS*, Pettersen MB*, . . . , **Lepperød ME**. *Coherently remapping toroidal cells but not grid cells are responsible for path integration in virtual agents*. iScience, 2023. Role: Senior/-corresponding author, supervised the project. Significance: Showed that AI agents spontaneously develop biologically consistent spatial cells, establishing our lab’s cognitive map expertise.
4. Pettersen MB, Schøyen VS, Malthe-Sørenssen A, **Lepperød ME**. *Decoding the cognitive map: Learning place cells and remapping*. eLife, 2024. Role: Senior/corresponding author. Significance: Demonstrated context-dependent memory through orthogonal remapping, key to FIND’s modular memory design.
5. Pettersen MB, Rogge F, **Lepperød ME**. *Learning place cell representations and context-dependent remapping*. NeurIPS, 2024. Role: Senior author. Significance: Extended cognitive map learning to context-switching, directly informing WP2’s compositional reasoning.
6. Kvalsund MK, Ellefsen KO, Glette K, Pontes-Filho S, **Lepperød ME**. *Sensor movement drives emergent attention and scalability in active neural cellular automata*. Neural Networks, 2026. Role: Senior/corresponding author. Significance: Demonstrated NCA-based inter-module communication with active vision—the primary orchestration mechanism for PAM.
7. Guichard E, Reimers F, Kvalsund MK, **Lepperød M**, Nichele S. *EngramNCA: A neural cellular automaton model of memory transfer*. ALife, 2025. Role: Co-supervisor. Significance: Showed NCA-based one-shot memory transfer between modules, directly enabling PAM’s distributed memory architecture.

8. Guichard E, Reimers F, Kvalsund MK, **Lepperød M**, Nichele S. *ARC-NCA: Towards developmental solutions to the Abstraction and Reasoning Corpus*. ALife, 2025. Featured on Machine Learning Street Talk podcast. *Role: Co-supervisor. Significance: Demonstrated abstract reasoning through developmental self-organization, informing WP2’s compositional reasoning.*
9. Schøyen VS et al. *Hexagons all the way down: Grid cells as a conformal isometric map of space*. PLOS Computational Biology, 2025. *Role: Senior/corresponding author. Significance: Provided geometric theory of neural representations, informing HDC integration in PAM.*
10. Danielli K, **Lepperød ME**. *A bio-inspired minimal model for non-stationary K-armed bandits*. Biological Cybernetics, 2026. *Role: Senior/corresponding author. Significance: Demonstrated bio-inspired adaptive learning under non-stationarity, directly relevant to WP3’s open-ended adaptation.*

Significant peer recognition

- **NORA Award for Early Career Investigator** (2025)—National recognition for contributions to NeuroAI
- **Simula ERC Incubator** (2022)—Selected for institutional support program for ERC candidates
- **Wharton Executive Education** (2022)—“High-Potential Leaders: Accelerating Your Impact”
- **Invited speaker:** University of California San Diego (2022), AI Week Oslo Met (2023), Keynote at Young Researchers in NeuroAI (2023), Simula Bio-inspired Computing Seminar (2024)
- **Workshop organizer:** 4-day NeuroAI workshop on Hurtigruten with 20 international speakers (2024); co-organizer of 3+ NeuroAI workshops connecting national and international labs (2023–2024)
- **Reviewer:** Nature Machine Intelligence, PLoS Computational Biology, eNeuro
- **Board member:** NORA Special Interest Group for NeuroAI Norway
- **Public outreach:** Brain Inspired podcast (2024), Machine Learning Street Talk podcast (2025), NRK interview (2021), opinion pieces in *digi.no* and *forskning.no*

Project leadership and supervision

Group leader, BioAI group at Simula (2020–present): 4.5 MNOK/year, managing 3 PhD students, 1 postdoc. Built the group from inception to a productive research unit with publications at NeurIPS, eLife, Science Advances, and Nature Communications.

Acting co-PI, Bio-inspired neural networks for AI application (2020–2026): 16 MNOK RCN-funded project, University of Oslo. Co-supervised 3 PhDs.

Supervised to completion: 2 PhD students (V. Schøyen, K. Holzhausen), 8+ MSc students (including graduates now at Stanford PhD program and UiO PhD program).

Research software and ML systems engineering

PAM / PAMNCA codebases: Large-scale ML engineering behind the preliminary results in this proposal: systematic GPU-cluster experimentation (~15K GPU-hours on A40s), automated result parsing from 150+ log files, reproducible evaluation pipelines, and benchmarking across 5 real-world domains (CRe50, MIT-BIH, ESC-50, VoxCeleb, embodied navigation) using official online continual-learning protocols. [TODO: repository URL and open-source release status]

Community research infrastructure: Co-developed NEST simulator modules within the Human Brain Project; co-developed the Exdir hierarchical data format and the Expipe experiment-management platform for reproducible neuroscience pipelines. [TODO: adoption indicators: downloads/stars/citing labs for Exdir and Expipe]

Benchmark contributions: Evaluation on official public protocols (CRe50-NI, ARC). [TODO: public leaderboard entries or benchmark submissions, if any]

Team profile to complement the PI: The FIND postdoc will be recruited from the ML systems community (target profile: PhD from an ICLR/ICML-publishing lab with large-scale training experience) to complement the PI’s neuroscience background and own the WP2–3 scaling infrastructure. [TODO: 1–2 sentences on candidate pipeline, e.g. named ELLIS/NORA contacts]

Additional information

Community leadership: Lead role establishing NeuroAI as focus field in the Norway–US Ministry of Education collaboration (2022). Developed and taught the NeuroAI course at University of Oslo (2023).

3 Part II of the Scientific Proposal

This section details the research methodology, work plan, and risk assessment for FIND. Part II complements Part I by providing the implementation specifics; it should not be read as a repetition of Part I but as a detailed elaboration of how the objectives will be achieved.

Research questions:

- Q1: How can memory-augmented modules distribute and combine long-term temporal information?
- Q2: How can such modules support active inference and causal reasoning?
- Q3: How can neurosymbolic components be integrated into distributed memory architectures without sacrificing gradient-based learning?
- Q4: Does this integration measurably enhance compositional reasoning in complex environments?
- Q5: Can such networks learn adaptively and reason about causal relations over long time horizons?

3.1 Research methodology

The core update equations underpin PAM’s mechanisms. Each module m has latent state $z_t^m \in \mathbb{R}^{d_z}$ and sensory input $x_t^m \in \mathbb{R}^{d_x}$ at time t . The prediction error $e_t = x_t^m - \hat{x}_t^m$ drives module state updates: $z_{t+1}^m = f_m(z_t^m, c_t^m, r_t^m, a_t)$, where f_m is a learnable function, a_t is the action, $r_c^m = \sum_i \alpha_c^m[i] h_i^m$ is context-dependent memory retrieval, and c_t^m is contextual input from inter-module communication. Gradients flow end-to-end through Gumbel-softmax routing and soft attention over memory slots on the read path; discrete WTA writes are handled with straight-through estimators (soft-read/hard-write), so encoder and modules train jointly while writes remain discrete. Two primary orchestration mechanisms will be explored:

1. **Message passing:** attention and convolution combined with aggregation—demonstrated in active vision [1]
2. **Neural cellular automata (NCA):** self-organizing principles for module communication—demonstrated in one-shot memory transfer [2] and abstract reasoning [3]

We commit to **NCA + message passing** as the primary orchestration mechanisms, supported by our PAMNCA results where NCA + distributed memory outperforms no-memory variants by 30–43%. Four further mechanisms form a structured ablation ladder in WP2: *Hypernetwork*, *Predictive routing*, *Competitive collaboration*, and *Energy-based orchestration*—ordered by increasing biological plausibility and implementation complexity.

WP1: Temporal Predictive Coding with Distributed Memory (Lepperød, PhD1; Q1; Years 1–3)

WP1 establishes the core PAM architecture for temporal prediction and memory. Building on our preliminary results (98.1% exact recall at $\sigma=0.4$, $3.75\times$ less forgetting than tPC-alone), WP1 extends the architecture to challenging temporal benchmarks and characterizes its operational regime.

Benchmark strategy. We adopt a four-tier evaluation hierarchy designed to systematically validate PAM’s capabilities and position it against the state of the art:

- **Tier 1 — Validation:** Synthetic tasks (Copy Task $T=100\text{--}5,000$; Adding Problem) and our existing MNIST/Omniglot benchmarks. These confirm core memory mechanisms work before scaling.
- **Tier 2 — Field positioning:** Standard time series forecasting benchmarks (ETT, Weather, ECL, Traffic) at horizons $\{96, 192, 336, 720\}$. Compared against DLinear, PatchTST [4], iTransformer [5], and Mamba using MSE/MAE metrics. These establish PAM’s competitiveness with purpose-built forecasting models.
- **Tier 3 — Core thesis:** Long-range benchmarks that test PAM’s distinctive claims. Sequential MNIST ($T=784$), Long Range Arena [6] including Path-X ($T=16,384$), Memory Maze [7] ($T=2,000\text{--}10,000$), and infinite dSprites [8] ($T=2,000\text{--}50,000$). If PAM achieves above-chance on Path-X, it demonstrates that HDC memory can compete with SSMs on ultra-long-range tasks.
- **Tier 4 — Continual learning:** Split MNIST/CIFAR (5 sequential tasks), concept drift benchmarks, and TSCIL [9] for time series class-incremental learning. These test whether PAM’s memory consolidation prevents catastrophic forgetting.

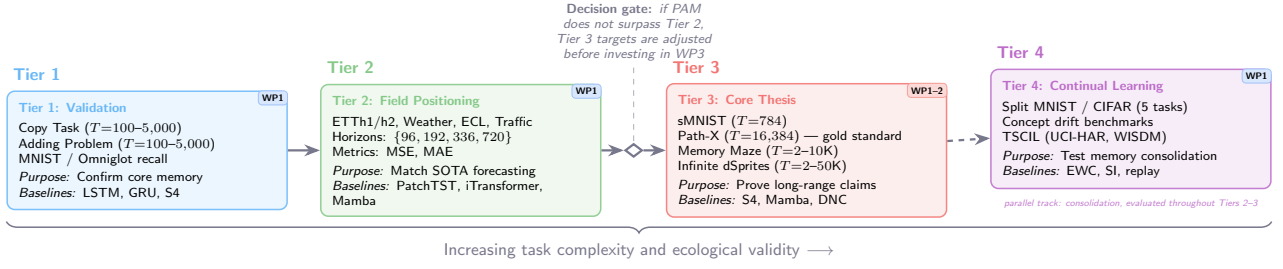


Figure 6: Four-tier benchmark hierarchy for PAM evaluation. Increasing sequence length, task complexity, and ecological validity from left to right. WP badges indicate which work package addresses each tier. A decision gate between Tiers 2 and 3 ensures incremental validation before committing to the most ambitious benchmarks.

T1.1: Temporal Benchmarks and Architecture Scaling

Completed: MNIST digit recall benchmark [10] (see Part I, Fig. 3). **Planned:** Validate on Tier 1 (Copy Task, Adding Problem) and Tier 2 (ETT, Weather, ECL) benchmarks. Visual encoding via ResNet [11]. Compare against GTMs [10], VRNNs [12], LSTMs [13], S4 [14], NTMs [15], and DNCs [16]. Offline probing [17] will analyze learned representations. *Success criterion:* PAM must achieve $\geq 90\%$ exact recall at $\sigma=0.3$ on Tier 1 tasks and MSE within 10% of PatchTST on Tier 2.

T1.2: Inference Depth, Noise Robustness, and Capacity

Systematically characterise the recall–noise–capacity landscape across $\sigma \in [0.0, 0.8]$ and $K \in [10, 500]$, including an analytical derivation of the recall cliff observed at $\sigma \approx 0.5$ from HDC crosstalk statistics (crosstalk in superposed hypervectors grows with stored items, predicting a capacity/SNR phase transition that neither predictive coding nor HDC theory predicts alone)—a falsifiable, architecture-specific prediction of PAM. Implement adaptive inference depth via prediction-error thresholding: at each iteration, compare the current prediction error against a learned threshold; terminate early when the error is below threshold, saving computation. Characterise interactions between slot count, HDC dimension, and forgetting rate under varying concept-drift speeds, providing design guidelines for WP2–3. Baselines include a memory-budget-matched episodic kNN/replay buffer—the store-everything alternative that also forgets nothing—to isolate PAM’s distinctive contribution: *selectivity* of what to store and retrieve under bounded memory. *Success criterion:* Adaptive inference must achieve $\geq 95\%$ of full-iteration accuracy at $\leq 60\%$ of compute on the MNIST benchmark, and PAM must match or exceed the budget-matched kNN baseline at equal memory footprint.

T1.3: Hierarchical Temporal Compression for Long-Horizon Memory

Extend PAM with hierarchical time-scale factorisation: fast modules ($\Delta t=1$) process step-level inputs while slow modules compress episodes into semantic summaries via learned gating inspired by complementary learning systems [18]. The gating criterion—whether and when prediction error triggers consolidation from episodic to semantic memory—will be resolved experimentally using prediction-error statistics from T1.2. T1.3 also develops *key reconsolidation*: periodically re-encoding stored memory keys with the current encoder so that content-addressable retrieval survives representation drift when the encoder learns online (the mechanism behind the 100% frozen-feature vs. 70.3% online-encoder gap on COrE50, Part I). Evaluate on Tier 3 benchmarks: “infinite dSprites” ($T=2,000-50,000$) and Memory Maze ($T=2,000-10,000$) [7]. The key test is **Path-X** ($T=16,384$): only S4, Mamba, and a few others solve this benchmark; success here would demonstrate that HDC memory with iterative inference can match purpose-engineered SSMs on ultra-long-range tasks.

WP2: Causal Reasoning and Compositional Understanding (Lepperød, PD; Q2–Q4; Years 2–4)

WP2 extends PAM with active inference, causal world model learning, and subspace-based compositional reasoning. This WP integrates neurosymbolic components for explicit reasoning alongside statistical learning, while enabling the system to discover causal structure through interventions.

T2.1: Action-Augmented Temporal Predictive Coding

Extend PAM to incorporate action nodes that modulate state transitions. Develop action-dependent prediction error computation across multiple timescales, including action-conditioned memory addressing, action-dependent update rules, and methods for maintaining temporal consistency. Design

joint optimization of state predictions and action selections. *Success criterion:* Action-conditioned prediction error must correlate $r > 0.7$ with observed state transitions in a controlled navigation task.

T2.2: Subspace Reasoning and Compositional Understanding

Develop mechanisms for compositional, subspace-based reasoning within the PAM architecture. The core idea is to factorize latent spaces into semantically meaningful subspaces that can be *composed algebraically* through HDC binding operations. Concretely, this means representing transformations as operations that compose exactly: a rotation followed by a colour change should produce the same result whether applied individually or as a single bound operation. Our preliminary PAMNCA results (24.7% exact match on ARC with 1.47M parameters) validate this approach, and memory ablations show that distributed memory contributes 30–43% of performance. Importantly, LoRA program decomposition experiments demonstrate that ARC programs are *not* low-rank composable, motivating the shift to HDC algebraic composition which operates in $d=1000+$ dimensional space with guaranteed compositionality. We will implement: (i) a lossless per-cell circular embedding for spatial structures; (ii) algebraically exact primitives (rotation, reflection, translation, colour transformations) that compose via HDC binding; (iii) a meta-learning pipeline (FOMAML [19]) for few-shot program inference from support examples; and (iv) HDC program storage enabling compositional search over the space of transformations. We evaluate compositional reasoning on abstract visual reasoning benchmarks including ARC [20], letter-string analogies [21], and bAbI multi-hop QA [22]. *Success criterion:* Extend PAMNCA from 24.7% to $>40\%$ exact match on ARC through HDC algebraic composition. Compositional retrieval must generalise to out-of-distribution rules (e.g., novel rotation–colour combinations). Our preliminary miniARC search achieves 100% OOD exact match on 4 novel rules. Figure 7 illustrates the approach.

T2.3: Causal World Model Learning through Active Inference

Implement active inference for action selection balancing information gain and goal-oriented reward [23, 24]. Expected free energy is made tractable by amortization: a learned policy posterior scores candidate sub-goals represented as HDC goal vectors, so planning operates in symbol space over bound sub-goal compositions rather than exhaustive state-space rollouts. Develop causal discovery mechanisms through interventionist exploration: the agent executes targeted interventions (changing one variable while holding others fixed), observes the downstream effects, and updates its world model to encode directional causal relationships rather than mere correlations. This follows the “interventionist” approach to causal learning [25]: modules learn not just that “opening a chest produces items” but that “the chest *caused* items to appear,” enabling counterfactual reasoning (“what if I hadn’t opened the chest?”). Build on our work with NCA [26, 1] and proximal policy optimization [27] for credit assignment. *Success criterion:* PAM with active-inference exploration must reach task completion $2\times$ faster than standard PPO on at least two benchmark environments.

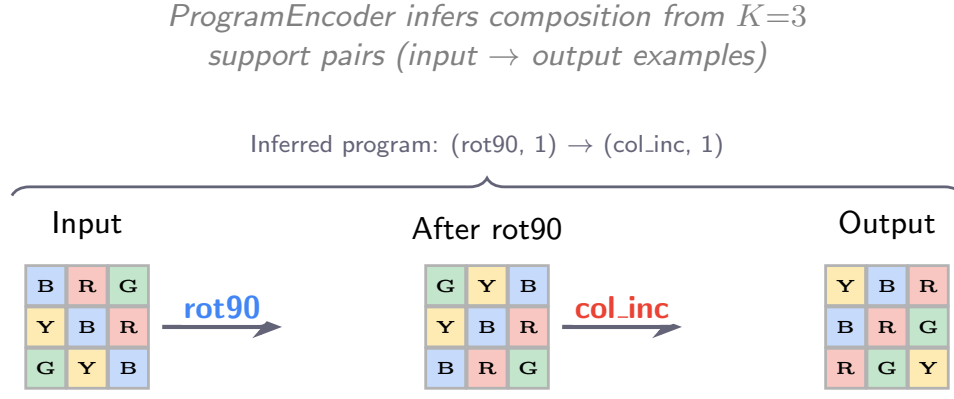
T2.4: Orchestration Ablation and Evaluation

Execute comprehensive ablation studies to quantify the contribution of each orchestration mechanism, distributed memory, and neurosymbolic component. Evaluate PAM on few-shot continuous control [28, 29] and long-horizon planning [7]. Investigate how PAM’s computations map onto neural circuits by extending our established RNN framework [30, 31, 32] that demonstrates spontaneous development of grid, place, and border cells. *Success criterion:* PAM must achieve >10 percentage point improvement over LSTM baseline on Memory Maze ($T > 500$ steps)—a deliberately intermediate gate; WP3 raises the bar to $T=10,000$ against stronger baselines (DQN, Dreamer). Orchestration ablations must show primary NCA/message-passing design outperforms at least two alternatives.

WP3: Long-Horizon Adaptation in Open-Ended Environments (Lepperød, PhD2; Q4–Q5; Years 3–5)

WP3 stress-tests PAM in environments demanding sophisticated planning, resource management, and adaptation to ever-changing dynamics. We adopt a tiered ambition structure:

- **Must achieve:** Memory Maze at $T=10,000$ with clear improvement (>10 pp) over LSTM/DQN baselines.
- **Should achieve:** Simplified Minecraft tasks (navigation, basic crafting) via Python Malmö API [33].



Algebraic composition in HDC space:

$$\underbrace{z_{\text{out}}}_{\text{output}} = \underbrace{\text{col_inc}}_{\text{colour LUT}} \circ \underbrace{\text{rot90}}_{\text{cell permutation}} (z_{\text{in}}) \equiv P_{\text{rot}} \cdot z_{\text{in}} \text{ then } \text{LUT}[\text{colour} + 1]$$

Out-of-distribution generalisation (zero-shot, no retraining):

Novel rule	Composition	Accuracy
rotate_270	rot90^3	100%
recolour_dec	col_inc^9	100%
flip_vertical	$\text{rot90}^2 \circ \text{reflect}$	100%
shift_down	$\text{rot90} \circ \text{translate} \circ \text{rot90}^3$	100%

Figure 7: Compositional reasoning through algebraic operations in HDC space. A visual transformation (rotation + colour increment) is decomposed into composable primitives. The ProgramEncoder infers the composition from support examples. Algebraic composition guarantees out-of-distribution generalisation to novel rules without retraining (bottom table; miniARC grids, 4 held-out rules).

- **Stretch goal:** Factorio production chains [34] requiring multi-step resource management across $T > 10,000$ steps.

T3.1: Environment Development and Curriculum Design

Design complex scenarios with varied challenges: resource management, crafting, building, combat, and exploration. Specific task designs include: (i) Memory Maze [7] at increasing horizons ($T=2,000$ – $10,000$); (ii) multi-step crafting chains in Factorio where intermediate products must be stored and recalled across $T > 10,000$ steps; (iii) spatial exploration in Minecraft where the agent must build a mental map of a procedurally generated environment. Create a procedural generation pipeline varying task complexity, resource constraints, and causal dependencies. Implement adaptive curriculum using automatic curriculum learning [35] and intrinsic motivation [36]: task difficulty increases only when the agent’s competence (measured via prediction accuracy) exceeds a threshold. Observations via frozen ResNet [11] encoder and game-state variables.

T3.2: Long-Horizon Active Inference Agent

Integrate PAM into an interactive agent with a three-level policy hierarchy: (L1) step-level motor primitives via PPO [27]—low-level actions like movement, item use, crafting; (L2) sub-goal selection via hindsight experience replay [37]—mid-level objectives like “go to location X,” “collect item Y”; (L3) strategic planning as symbolic goal vectors in HDC space [38, 39]—high-level plans like “build a shelter before nightfall” composed from bound sub-goal vectors. Each level maps to PAM modules operating at different temporal scales: L1 uses fast modules ($\Delta t=1$), L2 uses medium modules ($\Delta t=10$ –

100), and L3 uses slow modules ($\Delta t=100-1,000$) with persistent memory. Co-train through RL and self-supervised prediction objectives.

T3.3: Causal and Adaptive Learning Analysis

Assess causal learning through interventions (changing game rules, introducing disturbances), counterfactual reasoning (simulating alternative scenarios), and analyzing the agent’s ability to predict effects of its actions. Measure module specialization by analyzing activity patterns, stored information types, and task-specific contributions. Compare against DQN [40], PPO, RND [41], Dreamer [42], DINO-WM [43], and MuZero [44]. *Success criteria:* (i) PAM achieves >10 pp improvement over DQN on Memory Maze at $T>5,000$ steps. (ii) On Minecraft sub-tasks, PAM reaches task completion at least $1.5\times$ faster than PPO baseline. (iii) Intervention experiments demonstrate $>20\%$ accuracy in predicting novel action effects.

3.2 Work and time schedule

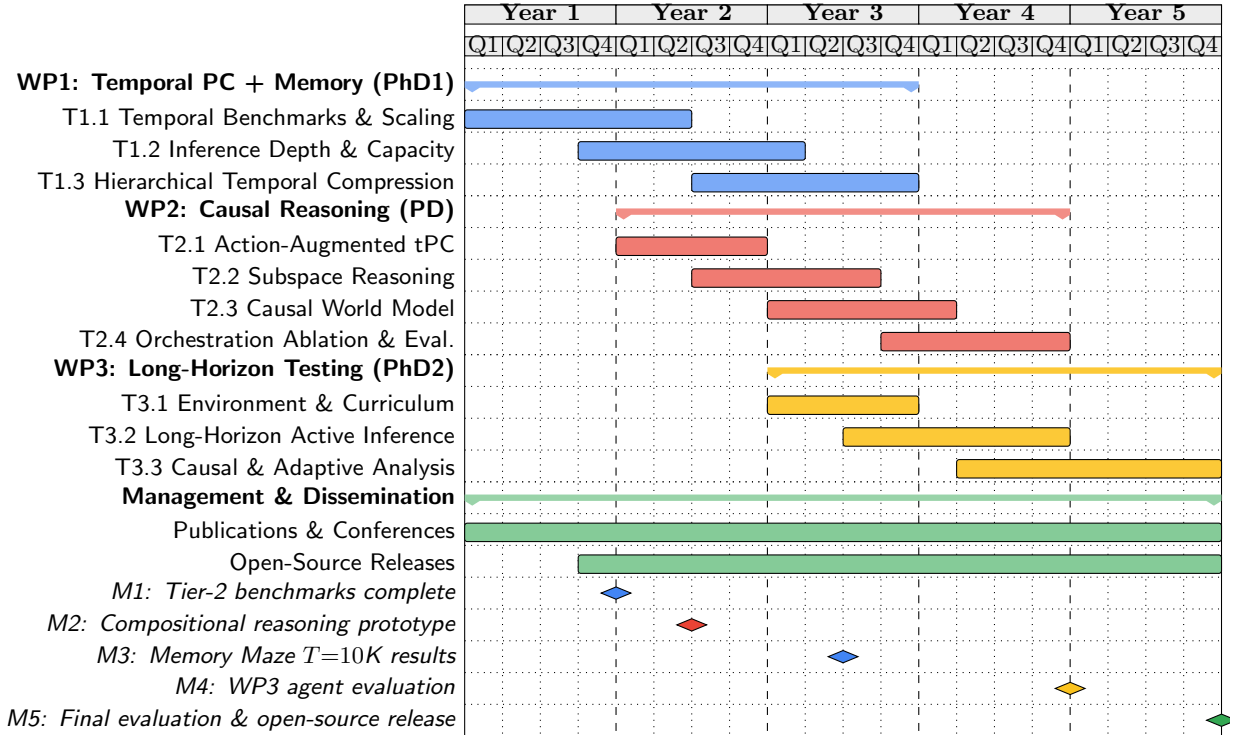


Figure 8: FIND project Gantt chart (60 months). WP1 (blue) Years 1–3, led by PhD1; WP2 (red) Years 2–4, led by PD; WP3 (yellow) Years 3–5, led by PhD2. Lepperød (PI, 50%) provides scientific leadership across all WPs.

Milestones:

- **M1** (Month 12): PAM Tier 2 benchmarks complete; CRe50-NI results published.
- **M2** (Month 18): WP2 compositional reasoning prototype; extend PAMNCA with HDC composition.
- **M3** (Month 30): Memory Maze results at $T=10,000$.
- **M4** (Month 48): WP3 environment integration and agent evaluation.
- **M5** (Month 60): Final evaluation, open-source release, and benchmark suite.

Estimated compute: WP1: $\sim 20K$ GPU-hours; WP2: $\sim 30K$ GPU-hours (informed by PAMNCA’s 15K hours for preliminary work); WP3: $\sim 80K$ GPU-hours (~ 800 GPU-hours per DreamerV3-scale training run $\times \sim 100$ experiments across curriculum stages and ablations). Total: $\sim 130K$ GPU-hours. Existing Simula eX3 and national Sigma2 allocations cover the WP1–2 development phase ($\sim 50K$ hours); the WP3 scale-up ($\sim 80K$ hours) exceeds these internal allocations and is what the additional funding request for compute covers.

Team allocation: Lepperød (PI, 50%) provides scientific leadership across all WPs and leads WP2. The Postdoc (PD, 100%) joins in Year 1, onboarding on WP1 infrastructure and benchmarks before WP2 tasks begin, then co-leads WP2 with Lepperød from Year 2, contributing to T2.1–T2.4. PhD1

(100%) starts Year 1 and leads WP1 (T1.1–T1.3). PhD2 (100%) starts Year 2 and leads WP3 (T3.1–T3.3). An additional PhD student may be hired in Year 3 depending on progress.

3.3 Feasibility, risks, and contingency plans

Cat.	Description	Like.	Sev.	Mitigation
Scientific	Modular architecture fails to specialize.	Med	High	Progressive training with soft inductive biases; Gumbel-softmax entropy floor prevents module collapse.
	Encoder drift degrades content-addressable retrieval (stored keys go stale as representations learn online).	Med	High	Key reconsolidation (T1.3); hybrid fast/slow memory with Bayesian consolidation; WTA write prevents slot mean-convergence.
	tPC-KM fails to scale beyond MNIST.	Med	Med	Start with simplified environments; 4-tier benchmark hierarchy ensures incremental validation.
	Compositional reasoning does not generalise beyond seen primitives.	Med	Med	Algebraic composition guarantees correctness by construction; FOMAML meta-learning for novel combinations.
Technical	Computational resources exceed expectations.	Med	Med	Sigma2 + eX3 allocations plus additional compute funding (Section 3.2); model distillation for efficiency.
	Orchestration bottlenecks in information flow.	Med	High	Structured ablation ladder provides 4 fallback mechanisms ordered by complexity.
	Active inference exploration is sample-inefficient.	Med	Med	Combine with imitation learning and adaptive curriculum; intrinsic motivation for efficient exploration.
Impl.	Environment integration (Minecraft/-Factorio).	Med	Med	Simplified environment wrappers with fallback to Memory Maze; established Python APIs available.
Project	Recruitment of specialized expertise.	Med	Med	Leverage NORA network, ELLIS community, and international collaborations for candidate identification.
Field	LLM-based agents partially achieve target capabilities during the project.	Med	Med	Value proposition shifts to efficiency, formal memory guarantees, and scientific understanding of memory; success criteria are efficiency-qualified (parameters, no pretraining).

Feasibility: The preliminary results (98.1% recall, $3.75\times$ less forgetting) demonstrate that the core architecture works. Each WP has independent value: even if WP3’s most ambitious targets are not fully reached, WP1–2 produce publishable architectures and benchmarks. The structured ablation ladder in WP2 ensures that at least one orchestration mechanism will succeed. The 4-tier benchmark hierarchy provides natural checkpoints: if PAM does not surpass Tier 2, the Tier 3 targets are adjusted before investing in WP3 environments.

3.4 Collaborations

Prof. Konrad Kording (University of Pennsylvania): Bayesian decision theory and causal inference (WP2). One co-authored publication, ongoing collaboration since 2018. *Tangible contribution:* Co-supervision of PD on causal world model formalism (T2.3); one PhD will visit Penn for 3 months in Year 3.

Prof. Marc de Kamps (University of Leeds): Computational neuroscience and compositional learning (WP2–3). Two co-authored publications, collaboration since 2016. *Tangible contribution:* Collaboration on mean-field analysis of modular dynamics; joint workshops on biological plausibility of orchestration.

Assoc. Prof. Blake Richards (Mila/McGill): Normative models of hippocampal activity. Co-PI, PhD exchange program, ongoing since 2023. *Tangible contribution:* PhD1 spends 6 months at Mila (Year 2) working on hippocampal models for WP1’s hierarchical memory; joint publications on normative models of memory consolidation.

Assoc. Prof. Arvind Kumar (KTH): Neuromodulators in AI. Co-PI and co-author, ongoing since 2020. *Tangible contribution*: Neuromodulatory gating mechanisms for memory consolidation (T1.3); joint supervision of PhD2.

Prof. Stefan Leutgeb (UC San Diego): Experimental spatial navigation. Visiting researcher 2017/2018, co-authored work submitted. *Tangible contribution*: Experimental validation of PAM’s spatial memory predictions against rodent place cell recordings.

3.5 Integration into host institution

Simula Research Laboratory is a Norwegian research institution with a strong track record in computational science. FIND will be hosted in the Department of Numerical Analysis and Scientific Computing (SCAN), within the BioAI group led by Lepperød. Simula provides:

- **Compute infrastructure**: Access to Simula’s eX3 HPC cluster (A100/H100 GPUs) and national Sigma2 (Saga, Olivia) resources, which cover the WP1–2 development phase; see the per-WP compute breakdown in Section 3.2 for the WP3 scale-up funded by the additional compute request.
- **Research environment**: Active AI research community, including the SURE-AI centre for industry collaboration, and connections to University of Oslo through Lepperød’s adjunct professorship.
- **Training programs**: Simula Academy for researcher development; NORA Research School for PhD training; established supervision track record (2 PhDs completed, 4 in progress).

3.6 Recruitment and ethics

Recruitment: PhD and postdoc positions will be advertised internationally through NORA, ELLIS, and Simula’s networks. We actively seek diverse candidates and will leverage our international collaborations for candidate identification. Simula’s competitive salaries and Oslo’s quality of life support strong recruitment.

Ethics: This research focuses on fundamental methods development in controlled game environments. No human subjects, animal data, or personal data are involved. We will follow Simula’s Code of Ethics, the EU Ethics Guidelines for Trustworthy AI, and the EU AI Act. Beyond the project, applications of FIND technology would require evaluation of biases, security, and data governance.

Funding ID

Funder	Amount	Subject	Status
Simula internal	4.5 MNOK/yr	BioAI group baseline funding	Active
RCN IKTPLUSS	12 MNOK	PASSION: AI safety and alignment	Submitted
RCN FRIPRO	10 MNOK	FIND (FRIPRO version): NeuroAI memory	Not funded
RCN IKTPLUSS	12 MNOK	constrAIIn: Constrained AI methods	Submitted

The proposed ERC project is distinct from all listed funding. The FRIPRO FIND application (3-year, smaller scope) was not funded; the ERC proposal represents a substantially expanded 5-year programme with WP2 (compositional reasoning, validated by PAMNCA on ARC), WP3 (open-ended environments), the postdoc position, and substantial preliminary results (CORE50 continual learning, ARC program synthesis, HDC compositional reasoning benchmarks) not available at the FRIPRO submission.

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